EE6083 VLSI Testing—Homework 3

1. **Scan Test**
   Figure 1 shows the circuit of a FSM. Complete the following questions.

   (a) Convert the circuit by adding scan to the three flip-flops. Create a complete scan test for the indicated faults (α, β, γ). Show the sequence of test vectors that are applied to this circuit in order to detect the faults, and show the sequence required to scan out and observe the results. (3 points)

   (b) Think an approach to reduce the switching power induced by the shifting operation of the scan without incurring extra hardware overhead. (3 points)

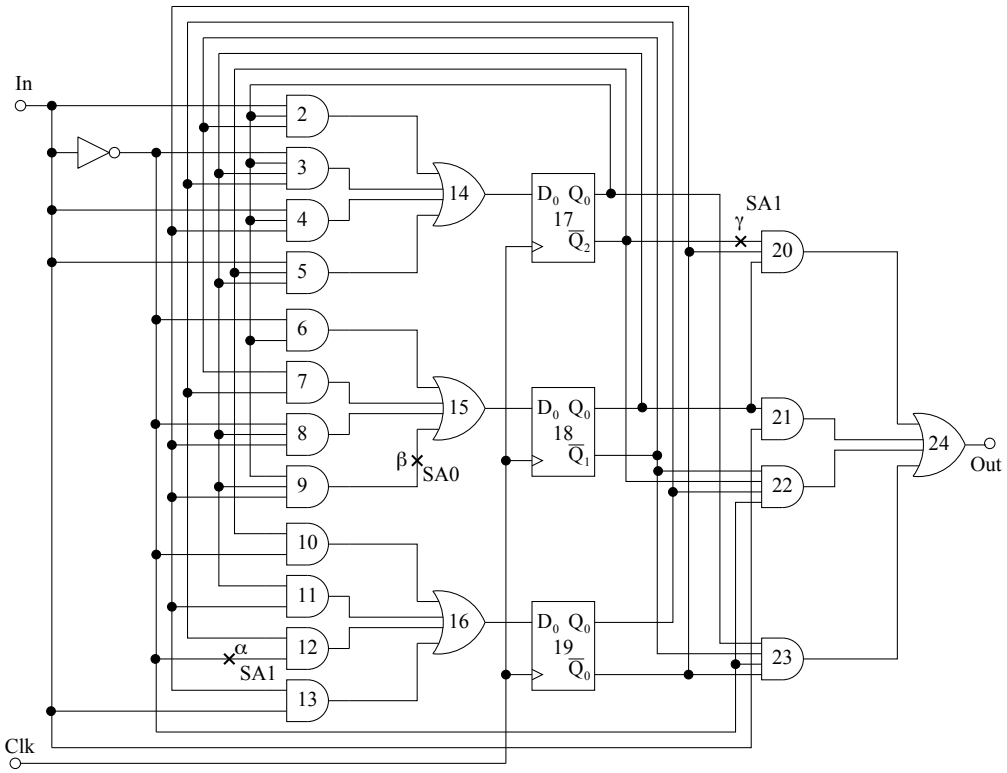


Figure 1: An example of a FSM.

2. **Scan Testing Time Reduction**
   A circuit with 1000 scan flip-flops has three uniquely identifiable blocks of logic. Block1 has 200 scan flip-flops and requires 300 scan vectors, block2 has 300 flip-flops and requires 80 scan vectors, and block3 has 500 scan flip-flops and requires 700 vectors.

   (a) If you could not break up the scan chains, how would you organize them to minimize test time? (2 points)

   (b) If you could break up the longest scan chain into scan chains of length 300 and 200, with each requiring 400 scan vectors, how would you organize the scan chains to minimize test time? (3 points)

3. **LFSR**
   Design a modified 6-bit LFSR to generate exhuastive test patterns, i.e., 64 different test patterns. Implement your design using the Verilog HDL and show the simulation waveforms to justify your design. (4 points)

1.(a)

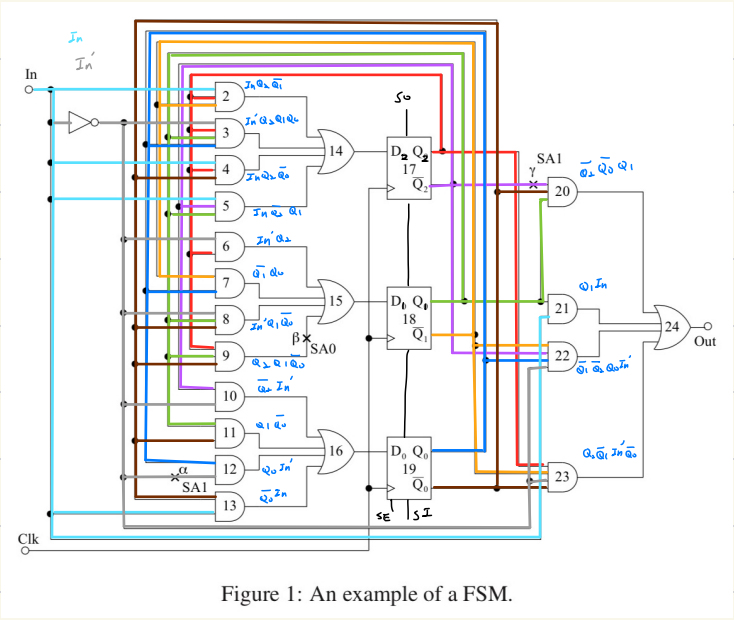


Figure 1: An example of a FSM.

α fault:

Active: $In = 1 \Rightarrow In' = 0$

Prop: $Q_0 = 1$

β fault:

Active: $\overline{Q_0} = 1 \Rightarrow Q_0 = 0$, $Q_1 = 1$, $Q_2 = 1$

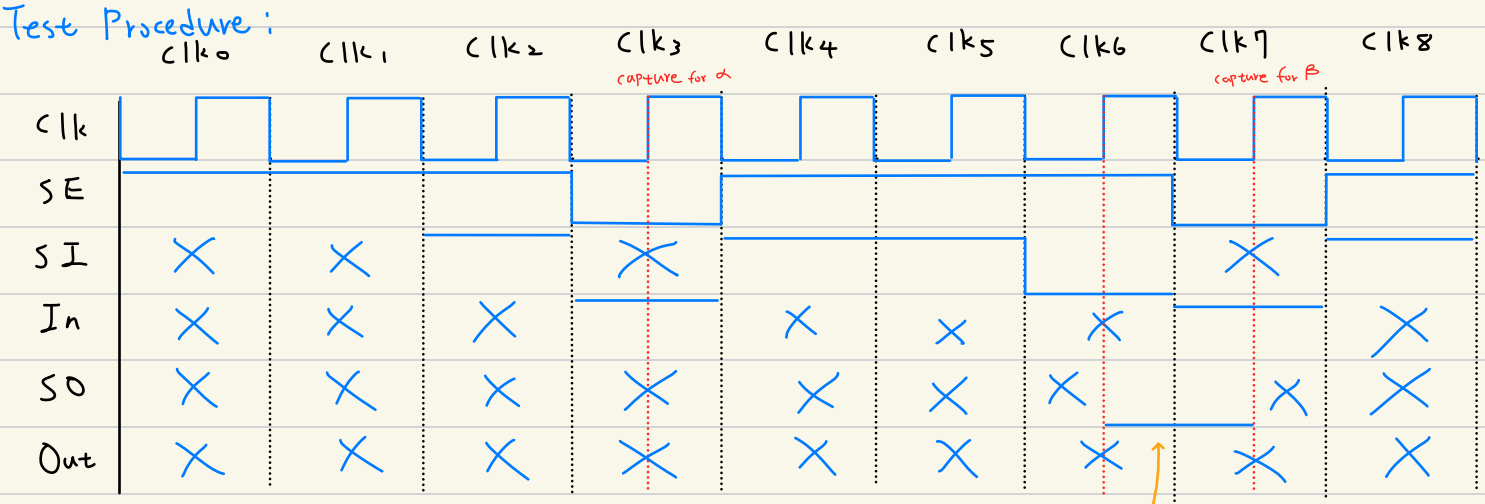
Prop: $In' = 0 \Rightarrow In = 1$

γ fault:

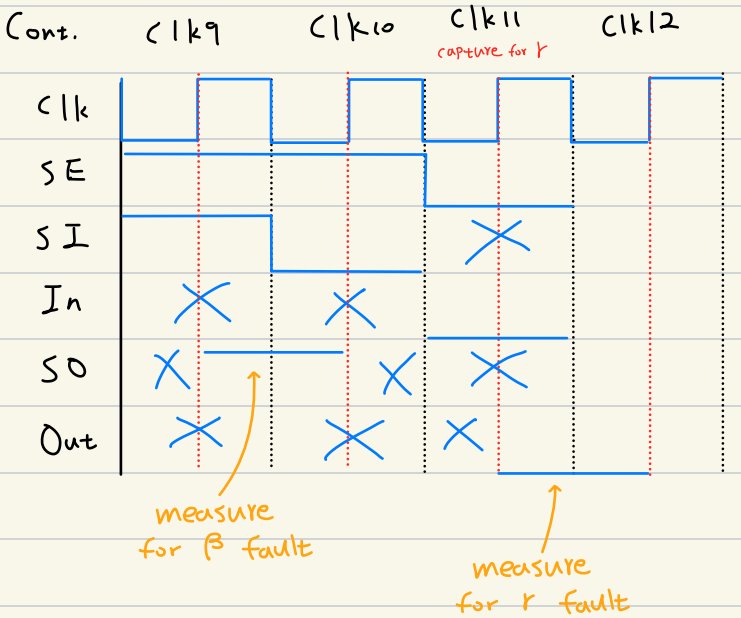
Active: $\overline{Q_2} = 0 \Rightarrow Q_2 = 1$

Prop: $\overline{Q_0} = 1 \Rightarrow Q_0 = 0$, $Q_1 = 1$, $In = 0$

Test Procedure:



新電路圖



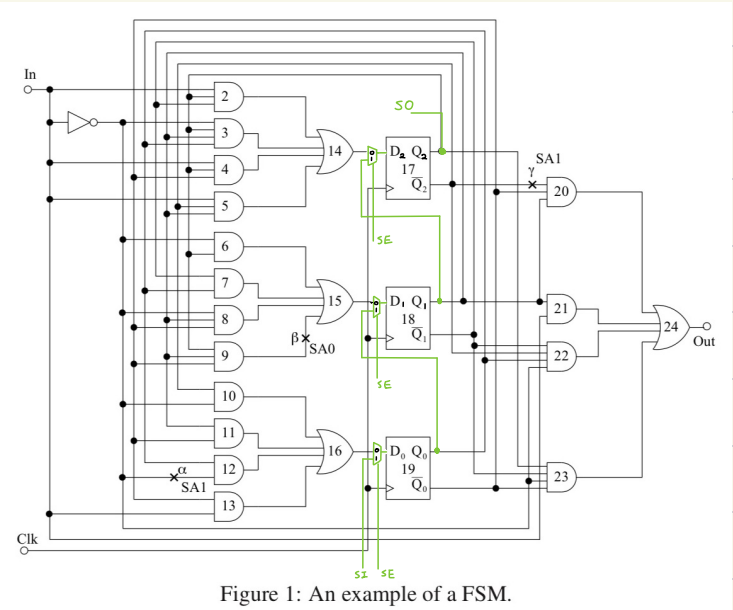


Figure 1: An example of a FSM.

note: X is don't care or random bits

一開始先測試 α fault，I_n 給 0($\bar{I_n}=1$) => Active fault，$Q_0=1$ => fault Prop.

所以 Clk2 時 SI 給 1($Q_0=1$)，然後 I_n 給 1，Clk3 做 capture，然後於 3 個 Clk 後(Clk 6) 可以觀察 SO 是否為 0

測試 β fault，$\{Q_2, Q_1, Q_0\}=\{1, 1, 0\}$ => Active fault，所以 Clk4、Clk5、Clk6 SI 依序給 1、1、0，然後 In 給 1 使 fault Prop.，Clk 7 做 capture，於 2 個 Clk 後(Clk9)，觀察 SO 是否為 1

測試 γ fault，Q_2 給 1($\bar{Q}=0$) => Active fault，$\{Q_1, Q_0\}=\{1, 0\}$，$I_n=0$ => fault Prop.

所以 SI 依序給 1、1、0 在 Clk8、Clk9、Clk10，然後 $I_n=0$，於 Clk11 capture，之後觀察 Out 是否為 0

1.(b) 為了減少 switching power induced by shifting operation of scan，如果可以避免 Shifting，消耗的 power 就會減少。

=> 每個 DFF 有自己的 SI、SO 訊號，i.e $SI=\{SI_0, SI_1, SI_2\}$，$SO=\{SO_0, SO_1, SO_2\}$，如右圖電路所示

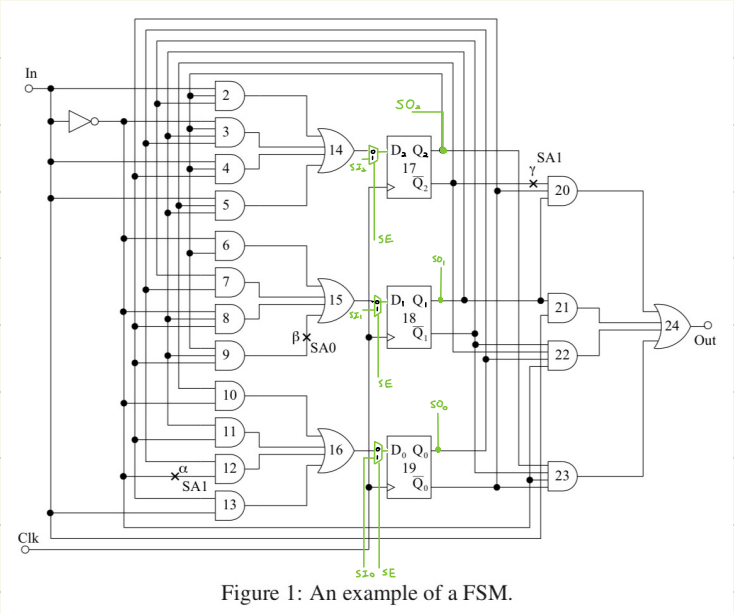
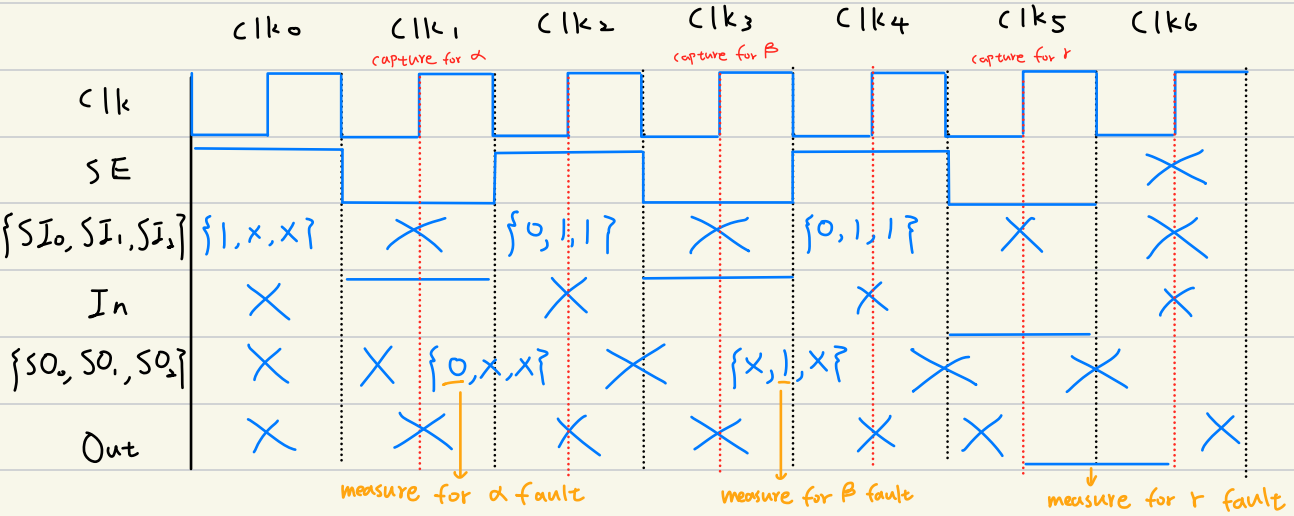


Figure 1: An example of a FSM.

New Test Procedure:



2. (a)

在只有2條 scan path 的情況下，將 block1 和 block2 接在一起(ie 同一條 scan path) 的情況下 才會有最少 test time

(1) block1 connect block2 test time : (300+80+1)(200+300)+300+80 = 190880 clock periods

(2) block3 test time : (700+1)(500)+700 = 351200 clock periods

=> minimize test time is 351200 clock periods，如果是 block3 跟其它 block 串聯，test time 只會更長

(b)

block1 : 200 n_{sff}, 300 n_{comb}

block2 : 300 n_{sff}, 80 n_{comb}

block3 : 500 n_{sff}, 700 n_{comb} → block3-1 : 300 n_{sff}, 400 n_{comb}; block3-2 : 200 n_{sff}, 400 n_{comb}

將 block1 和 block3-2 接一起，block2 和 block3-1 接一起

(1) block1 connect block3-2 test time : (300+400+1)(200+200)+300+400 = 281100

(2) block2 connect block3-1 test time : (80+400+1)(300+300)+80+400 = 289080 clock periods

=> minimize test time is 289080 clock periods

3.

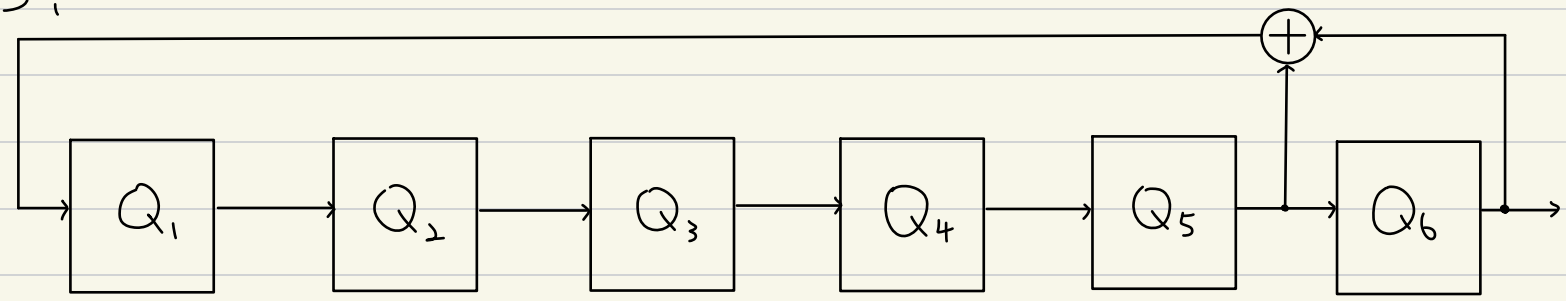


Fig 3-1、6 bit LFSR structure

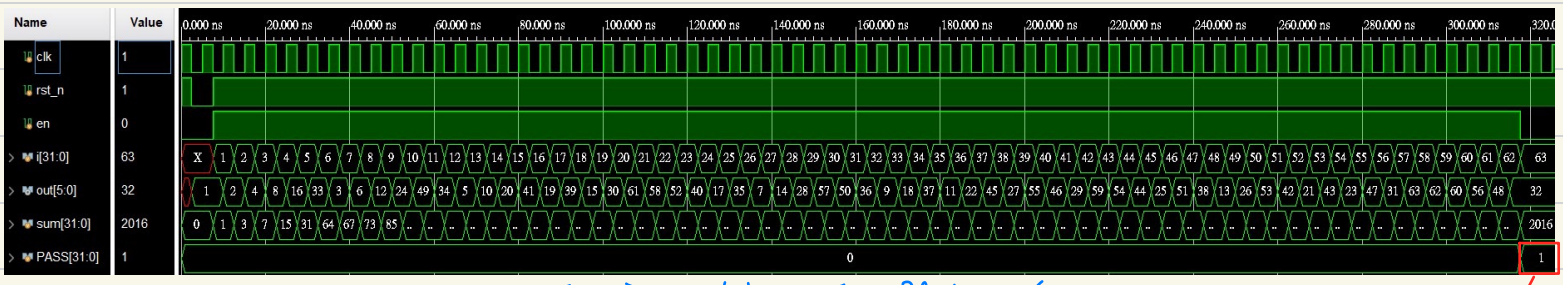


Fig 3-2、6 bit LFSR PASS waveform

PASS = 1, 6 bit LFSR 成功通過測試